

Instructions to Candidates

- Full marks for correct answers to all 3 questions.
- Provide answers in a single .pdf document (exported from Word/Scan of Handwritten material/ etc). Start each question on a separate page.
- Put all R code used into a single .R script file. Clearly separate using comment lines individual parts of the script. Save work regularly.

Question 1 [30 marks]

The data in DigitsF.Rdata is a $30 \times 1000 \times 10$ array of 30 features derived from the 1000 sample handwritten images of each of the 10 digits in the Digits.Rdata data set that we used in class. Evaluate the hypothesis that the population mean of these features are the same *i.e.* $H_o : \mu_0 = \mu_1 = \dots = \mu_9$. Under the Null Hypothesis, assume that feature values have a common, but not necessarily Gaussian, multivariate distribution across all digits.

- (a) Write an R script that evaluates the test-statistic

$$Test = \sum_{k=0}^9 (\hat{\mu}_k - \hat{\mu})' \hat{\Sigma}^{-1} (\hat{\mu}_k - \hat{\mu})$$

where $\hat{\mu}_k$ is the mean of the 1000 feature vectors available for the k 'th digit and $\hat{\Sigma}$ is the covariance matrix for each digit, averaged across digits.[4]

- (b) Report the value of the test statistic for the data.[3]
- (c) Write an R script to simulate, using a suitable **bootstrapping process**, $NS = 1000$ values of the test statistic in (a) under the Null Hypothesis. **Be sure to have `set.seed(101)` as the first line of the script.**[4]
- (d) Run your procedure and report the first 3 simulated values of the test statistic obtained. Use the bootstrap-generated null distribution to evaluate the p-value associated with the test value obtained on the data. State this p-value. [4]
- (e) Write an R script to simulate, using a suitable **permutation process**, $NS = 1000$ values of the test statistic in (a) under the Null Hypothesis. **Be sure to have `set.seed(101)` as the first line of the script.**[5]
- (f) Run your procedure and report the first 3 simulated values of the test statistic obtained. Use the permutation-generated null distribution to evaluate the p-value associated with the test value obtained on the data. State this p-value. [5]
- (g) Create a suitable plot to compare the bootstrap and permutation generated null distributions and provide an interpretation of what the plot shows. [5]

Question 2 [30 marks]

Consider the USArrests data in R. It is of interest to examine the hypothesis that variables are uncorrelated, i.e. that the correlation matrix (R) of the 4-variables is the identity matrix. The negative logarithm of the determinant of the correlation matrix has been proposed as a suitable test statistic - i.e. $\hat{\lambda} = -\log(|\hat{R}|)$. Excessively large values are rejected.

- (a) Report the value of the test statistic [2].
- (b) Evaluate the Bootstrap distribution of the test-statistic- use 5000 bootstrap samples for this.

Report the bootstrap estimate of the standard error in $\hat{\lambda}$. - [2].

Report your result in a density scale histogram. Include on your diagram a fit of a Gamma distribution to the histogram data (using the `fitdistr()` function in R). Your answer should be reported in a plot with suitable labelling and legend - [8].

- (c) By direct simulation evaluate the Null distribution for the test statistic under the assumption that the Null hypothesis the data are a random sample from an underlying multivariate normal distribution. Use 5000 replication for your simulation.

Report your result in a density scale histogram. Include on your diagram a fit of a Gamma distribution to the histogram data (using the `fitdistr()` function in R). Report a suitable plot with your results - [7].

Is the Null hypothesis rejected at a 1% level? [1].

- (d) Use a suitable Bootstrap simulation procedure (again based on 5000 simulations) to evaluate the distribution for the test statistic under the Null hypothesis assumption that, after centring variables by the true means and scaling by the true standard deviations, the transformed data matrix can be viewed as a random sample from an underlying multivariate distribution with iid components.

Report your result in a density scale histogram. Include on your diagram a fit of a Gamma distribution to the histogram data (using the `fitdistr()` function in R). Report a suitable plot with your results - [7].

Is the Null hypothesis rejected at a 1% level? [1].

- (e) Formally compare the two Null distributions using the two-sample Kolmogorov-Smirnov test, `ks.test()`, in R.[2]

Question 3 [40 marks]

- (a) Let X be a p -dimensional random variable. Provide a detailed proof of the formula

$$\text{Cov}(X_j, X_k) = E[X_j X_k] - EX_j EX_k \quad , \quad 1 \leq j, k \leq p \quad [\mathbf{10}]$$

- (b) Suppose X_1 and X_2 are two p -dimensional multivariate random variables - not necessarily independent. Provide formulas for the mean and covariance of the variable of the variable $Y = X_1 + X_2$ **[10]**
- (c) Suppose $X_1 \sim N_p(\mu_1, \Sigma_1)$ and $X_2 \sim N_p(\mu_2, \Sigma_2)$ are independent. Use characteristics functions to prove that $Y = X_1 + X_2$ is MVN **[10]**.
- (d) The binary image file `cald0814Attn.F01.floatImage` is a crude 3-d tissue density brain scan. The data are an array of 4-byte real numbers - 35 planes of data with each image plane consisting of 128x128 pixels. Note pixel dimensions are 2.25x2.25mm and the slice thickness is 4.5mm. Use `readBin()` in R to read the the data and use the `image` function to create images showing the mid- volume orthogonal slices (transverse, coronal and sagittal) through the data volume. Also create a histogram of the tissue density values (after values outside of the brain are excluded). Assemble the 4 plots in a suitably labeled a (2x2) array with a legend. **[10]**